GGG-B-001351 (GSA FSS)
March 28, 1968

INTERIM FEDERAL SPECIFICATION

BROACHES, JEWELER'S AND PIVOT

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers jeweler's broaches for cleaning, reaming, or enlarging holes in jewelery, instruments, etc.

1.1.1 Federal specification coverage. Federal specifications do not cover all types, styles, and sizes of the commodities as implied by the specification titles, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and styles. Broaches covered by this specification shall be of the following types and styles:

Type I. Square tang (jeweler's).
  Style 1. Pentagon blade.
  Style 2. Round blade.

Type II. Round tang (jeweler's pivot).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 5110

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Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-P-17667 - Paper, Wrapping, Chemically Neutral.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3.1 Illustrations. The illustrations shown herein are for the purpose of identification and are not intended to preclude any broach which otherwise meets the requirements of this specification.

3.2 Material. The broaches shall be made of high quality steel, and suitably designed for the purpose intended.

3.2.1 Blade. The cutting or polishing surfaces and blade length shall be designed in accordance with the manufacturer's standard practice, and hardened to a Rockwell "C" 60 to 65 to meet the test, as specified in 4.5.1.

3.2.2 Tang. Tangs shall be "drawn", or not hardened, and suitably shaped and sized for use in a pin vise.

3.2.3 Marking. Individual broaches shall be marked with the corresponding size, and cases or containers shall be so marked that the source of manufacture is readily identified.

3.3 Type I. Square tang (jeweler's).

3.3.1 Type I, style 1, pentagon blade. Type I, style 1 broaches shall have a tapered, five-cornered blade. The largest broaching section of this blade shall be either stubs standard wire gage, or twist drill size and shall be furnished in size 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 52, 55, 60, 65, 70, 75, or 80, as specified.

3.3.1.1 Style 1, broaches shall be similar to figure 1 and of one of the sizes specified in 3.3.1 (see 6.2).

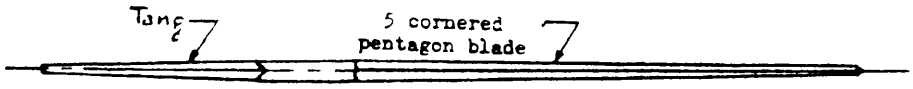


FIGURE 1. Type I, style 1, pentagon blade.

3.3.2 Type I, style 2, round blade. Type I, style 2 broaches shall have a tapered, round blade. The largest section of this blade shall be stubs standard wire gage size and shall be furnished in size 15, 20, 25, 30, 35, 40, 50, 55, 60, 65, or 70, as specified.

3.3.2.1 Style 2, broaches shall be similar to figure 2 and of one of the sizes specified in 3.3.2 (see 6.2).

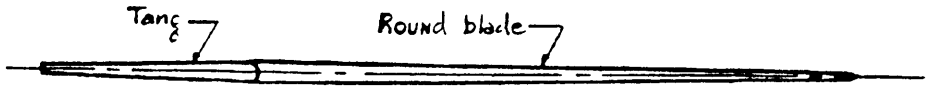


FIGURE 2. Type I, style 2, round blade.

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3.4 Type II

3.4.1 Type II, round tang (jeweler's pivot). Type II broaches shall have a round tang, and tapered cutting blades designed to produce holes within the range limits specified for the following sizes, as specified:

Size 1, 0.003 to 0.013; size 2, 0.005 to 0.015; size 3, 0.009 to 0.0625; and size 4, 0.053 to 0.125.

3.4.1.1 Type II broaches shall be similar to figure 3 and of one of the sizes specified in 3.4.1 (see 6.2).



FIGURE 3. Type II, round tang (jeweler's pivot).

3.5 Workmanship. Workmanship shall be of the highest grade throughout, and equal in every respect to good commercial practice. Broaches shall be free from imperfections that may effect either their appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table I.

TABLE I. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual and dimensional examination	1 ea.	II	4.0	Defects per hundred units	4.3.1
Testing	1 ea.	S-3	1.0	Percent defective	4.5
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.6

4.3 Examination

4.3.1 Visual and dimensional examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, marking, and dimensions.

4.4 Classification of defects. Inspection of the broaches shall be conducted as follows:

4.4.1 Visual and dimensional examination.

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Categories and Defects

AQL (approx.) 1.5 percent

100. Critical:
None defined.

200. Major (sampling per table I).
200.1 Type, style, and size not as specified.
200.3 Materials not free of cracks, splits, or breaks.
200.4 Evidence of rust, fins, burrs, or rough edges.

4.5 Tests.

4.5.1 File hardness test. The hardness of broaches shall be determined by the use of two special test files, one having a Rockwell "C" hardness of 58/59, and the other 66/67. The broach to be tested shall be rigidly mounted in a suitable clamping device, and both test files drawn across the largest working area of the broach. The 58/59 file should glide across the broach's surface without penetration. The 66/67 file should bite into the broach's surface. If these surface conditions exist, the broach shall be considered as meeting the hardness test.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any components missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The broaches shall be cleaned by process C-3 and dried by procedure D-1 in accordance with MIL-P-116.

5.1.1.2 Unit packaging. Each broach shall be packaged in accordance with method IC-1 of MIL-P-116.

5.1.1.3 Intermediate packaging. Unless otherwise specified, 12 unit packages of like description shall be packaged together in a close fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 Level B.

5.1.2.1 The broaches shall be cleaned by process C-3 and dried by procedure D-1 in accordance with MIL-P-116.

5.1.2.2 Unit packaging. Twelve broaches of like description shall be wrapped together with material conforming to MIL-P-17667 and packaged in a box conforming to PPP-B-566, style II, type A with pye locks, class a. The box shall be closed in accordance with the appendix to the box specification.

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5.1.2.3 Intermediate packaging. Seven unit packages of like description shall be packaged together in a box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The box shall be closed in accordance with the appendix to the applicable box specification.

5.1.3 Level C. The broaches shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Broaches of like description packaged, as specified in 5.1.1.3, shall be packed in a close fitting box conforming to PPP-B-636, class 2, style RSC. The gross weight shall not exceed the weight limitations of the box specification. The box shall be closed and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Six intermediate packages of like description packaged, as specified in 5.1.2.3, shall be packed in a close fitting box conforming to PPP-B-636, class 1, style RSC. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The broaches shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to marking required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style, and size required (see 3.3.1, and 3.3.2).
(c) Selection of applicable level of preservation and packaging and packing required (see 5.1 and 5.2).

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